PROKUDIN-GORSKII PROJECT REPORT

INTRODUCTİON

Prokudin-Gorskii was a pioneer and genius of early color photography. He captured the same scene consecutively using three different color filters (Blue, Green, and Red). However, due to the physical movement of the camera and the time elapsed between each exposure, these three separate channel images are not perfectly aligned with each other. When these images are simply superimposed, it results in noticeable color shift problems.

Alignment Metric Selection

During the alignment phase, the goal is to measure the correspondence between the shifted (moving) and fixed (reference) images. To achieve this, various similarity metrics are utilized (such as SSD, SAD, and NCC).

For this project, I chose to implement the **SSD (Sum of Squared Differences)** metric. The primary reasons for this selection are:

1-Efficiency

2-Speed

3-Compatibility

4-Accuracy of Results

How SSD works: It subtracts the corresponding pixel values of the two images, squares the differences, and sums them up. The smaller the resulting value, the higher the similarity between the images.

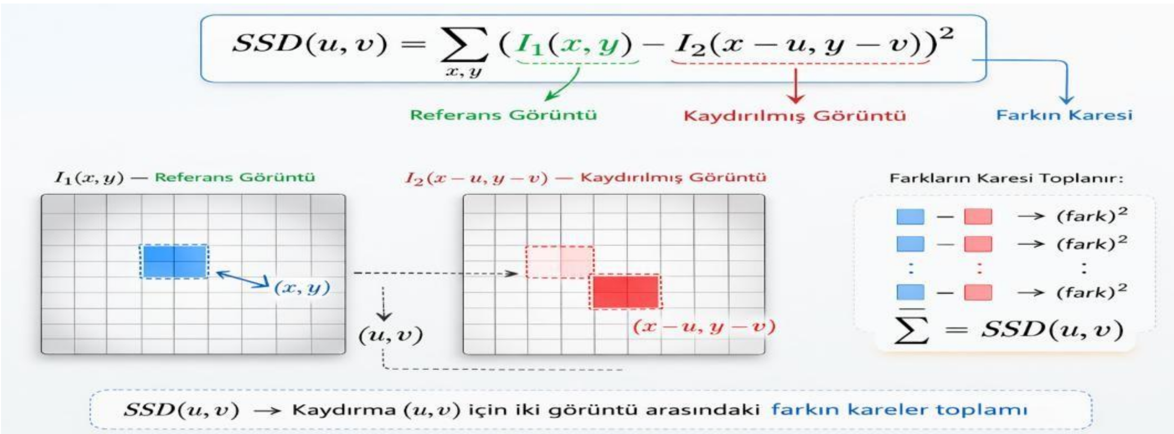

Table of Alignment Displacements

PROJE SONUÇ RAPORU VE HİZALAMA DEĞERLERİ

Görsel İsmi	Yeşil (G) Vektör	Kırmızı (R) Vektör
00056v.jpg	(6, 1)	(13, 1)
00088v.jpg	(3, 3)	(4, 5)
00106v (1).jpg	(4, 1)	(9, -1)
00106v (2).jpg	(4, 1)	(9, -1)
00106v (3).jpg	(4, 1)	(9, -1)
00106v.jpg	(4, 1)	(9, -1)
00153v.jpg	(7, 3)	(-15, 15)
00163v.jpg	(-3, 1)	(-4, 1)
00194v.jpg	(4, 2)	(8, 4)
00398v.jpg	(5, 3)	(11, 4)
00458v.jpg	(4, 1)	(9, 3)
00600v.jpg	(6, 4)	(13, 6)
00757v.jpg	(2, 3)	(5, 5)
00804v.jpg	(6, -2)	(13, -4)
00888v.jpg	(6, 1)	(12, 0)
00889v.jpg	(1, 2)	(4, 3)
00907v.jpg	(3, 1)	(6, 0)
00911v.jpg	(1, -1)	(13, -1)
01031v.jpg	(1, 1)	(4, 2)
01164v.jpg	(6, 2)	(11, 3)
01167v.jpg	(5, 0)	(12, -2)
01269v.jpg	(6, 2)	(14, 3)
01522v.jpg	(6, 2)	(13, 2)
01597v.jpg	(7, 1)	(15, 1)
01598v.jpg	(8, 0)	(15, -1)
01728v.jpg	(8, 1)	(15, 1)
01880v.jpg	(6, 2)	(14, 4)
10131v.jpg	(6, 2)	(12, 3)
31421v.jpg	(8, 0)	(13, 0)

Toplam İşlenen Görsel: 29

Upon analyzing the data in the table, it is observed that the displacement of the Red channel is approximately twice that of the Green channel. This observation provides insight into the physical structure of the camera used by Prokudin-Gorskii, allowing us to make deductions about its fundamental characteristics. Typically, the filter arrangement on the glass plates is ordered as Blue (Top) -> Green (Middle) -> Red (Bottom). Since our reference point is the Blue channel at the top, the physical distance to the bottommost Red frame is the greatest, resulting in a larger displacement.

SAMPLE INPUTS AND OUTPUTS OF THE PROJECT

INPUT

UNALIGNED

ALIGNED

ENHANCED

INPUT

UNALIGNED

ALIGNED

ENHANCED

INPUT

OUTPUT

Challenges Encountered

During this project, I primarily encountered two main challenges:

Image Cropping (Border Artifacts): The black scanning borders and physical damages at the edges of the initial raw images caused the alignment algorithm (SSD) to focus on the wrong points. To resolve this issue, I applied a 7% crop to the image edges, thereby obtaining cleaner data for processing.

Motion-Induced Blur: Due to Prokudin-Gorskii's shooting technique, capturing the three different channels at slightly different times led to movement within the scene, resulting in motion blur.

CONCLUSION

Throughout this project, the digital restoration and automatic colorization of the early 20th-century Sergey Prokudin-Gorskii glass plate collection were successfully achieved after extensive work. The algorithm developed in Python processes the raw black-and-white negatives by taking the Blue channel as the base reference and optimally overlapping the other channels using various alignment methods.

Takeaways:

Noise Management: Efforts were made to improve alignment success by addressing border noise encountered during processing. Consequently, this problem was effectively solved by applying a 7% crop from the image edges, significantly enhancing the final visual quality.

Algorithmic and Analytical Thinking: Throughout the project, I had the opportunity to improve my skills by applying many concepts and techniques learned in the Computer Vision course to a real-world problem. This project contributed significantly not only to my understanding of computer vision but also to broadening my general engineering perspective, particularly regarding Python programming, project organization, and structural design.

Academic Achievement: Successfully completing such a comprehensive and practical project directly demonstrates my proficiency and dedication in this field, contributing significantly to my academic career.

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